

FinWise AI: Intelligent Loan Eligibility, Credit Analysis & Financial Advisory Platform

Complete Technical Documentation

A comprehensive web-based intelligent financial advisory system combining Artificial Intelligence, cloud computing, and financial analytics to assist users in making informed financial decisions.

Technologies: HTML, CSS, JavaScript, Claude AI, Google Sheets, Google Apps Script

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Git Hub Repositories Link : <https://github.com/JinkaGovardhan/FinWise-Loan-AI>

Git Hub Live Website Link : <https://jinkagovardhan.github.io/FinWise-Loan-AI/>

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1.1 Introduction

The rapid advancement of Artificial Intelligence (AI) and Financial Technology (FinTech) has significantly transformed the banking and financial services industry. Traditional financial services such as loan approval, credit analysis, and financial planning often involve lengthy procedures, manual verification, and multiple interactions with financial institutions.

The FinWise AI platform is designed as a modern web-based financial application that combines Artificial Intelligence, cloud computing, and financial analytics to assist users in making informed financial decisions. Before beginning implementation, it is essential to establish a well-prepared development environment.

1.2 Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i5 (10th Gen+)	Intel Core i7 / AMD Ryzen 7
RAM	8 GB	16 GB
Storage	20 GB SSD	50 GB+ SSD
Monitor	Full HD (1920×1080)	Full HD or higher
Internet	Broadband connection	High-speed broadband

1.3 Software Requirements

IDE: Visual Studio Code (VS Code)

- Lightweight and extensible
- IntelliSense and syntax highlighting
- Integrated terminal and Git integration
- Extension marketplace

Browser: Google Chrome

- Advanced Developer Tools
- Network monitoring and debugging
- JavaScript debugging capabilities

Version Control: Git and GitHub

- Distributed version control

- Remote repository management
- Collaboration and deployment

Additional Tools:

- Live Server extension for automatic browser refresh
- Postman for API testing
- Node.js (optional for package management)
- Google Drive for cloud document management

1.4 Programming Languages and Technologies

Technology Purpose

HTML5 Structural foundation for web pages, forms, and dashboards

CSS3 Visual appearance with glassmorphism design, responsive layouts, animations

JavaScript (ES6+) Business logic, financial calculations, API communication, DOM manipulation

Claude AI API Natural language processing and financial recommendations

Google Sheets Cloud database for record storage

Google Apps

Script Middleware for secure data storage

1.5 Artificial Intelligence Requirements

Claude AI API:

- Advanced natural language processing
- Financial recommendation generation
- Context-aware analysis
- Personalized financial advice

Requirements:

- Anthropic account creation
- Valid Claude API key
- Secure API communication (HTTPS)
- Prompt engineering expertise

Security Considerations:

- API keys stored securely (not in frontend code)

- Environment variables for credentials
- Secure backend services for API calls

1.6 Cloud Database Requirements

Google Sheets:

- Lightweight cloud database
- Automatic backup and synchronization
- Accessibility and simplicity
- Minimal infrastructure requirements

Google Apps Script:

- Secure middleware for data storage
- HTTP request handling
- Spreadsheet row management
- Cloud synchronization

1.7 Security Requirements

Input Validation:

- Validate all user inputs before processing
- Reject invalid or malicious data
- Implement field-level validation

Data Protection:

- HTTPS encryption for all communications
- Secure API key management
- Input sanitization and output encoding
- Exception handling and error reporting

Access Control:

- Authentication mechanisms
- Authorization verification
- Secure permission management
- Rate limiting for API requests

1.8 Technical Knowledge Required

Web Development:

- HTML, CSS, JavaScript proficiency

- Responsive web design
- DOM manipulation
- Asynchronous programming (Promises, async/await)

Financial Concepts:

- Loan eligibility criteria
- EMI calculation formulas
- Credit score classification
- Debt-to-income ratio analysis
- Interest calculation methods
- Financial risk assessment

Cloud and APIs:

- REST API concepts
- JSON data structures
- Cloud computing basics
- Google Apps Script
- Git version control
- API integration techniques

1.9 Importance of Pre-requisites

Development Benefits:

- Reduces development complexity
- Improves software reliability
- Simplifies debugging and troubleshooting
- Ensures seamless technology integration

Quality Assurance:

- Prevents configuration problems
- Improves software maintainability
- Enhances code scalability
- Reduces implementation risks

Project Success:

- Enables efficient development
- Supports future enhancements

- Facilitates team collaboration
- Improves overall project reliability

1.10 Chapter Summary

This chapter discussed all essential prerequisites for developing the FinWise AI platform:

- Hardware configuration for optimal performance
- Software tools and development environment setup
- Programming languages and technologies selection
- Artificial Intelligence API requirements
- Cloud database and storage solutions
- Security measures and best practices
- Technical knowledge and skill requirements
- Importance of proper preparation

Establishing these prerequisites before coding begins ensures efficient development, secure financial processing, reliable AI integration, and scalable application architecture